Stress-Testing Geometric Interpretability Claims in Grokking

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Abstract

In this paper we test whether Hodge decomposition gives a consistent account of how a transformer's learned representations change as it groks modular addition. The generalising solution to modular addition has an independently understood Fourier structure, giving us a known representation against which to compare our geometric measurements. We measure how the same probe activations move between checkpoints and separate this motion into exact (gradient-like), coexact (curl-like), and harmonic components. The baseline run suggests coexact-led motion during the transition followed by exact-led motion after grokking, but this pattern does not appear in every run. The consistent result is a relative shift towards exact motion. From the transition to the post-grokking phase, the difference between the coexact and exact fractions decreases across all 56 combinations of eight training seeds and seven analysis settings, with uncertainty calculated across the eight independent seeds. At the baseline setting, the mean change is -0.267, with a 95% t interval of $[-0.400, -0.135]$. The result remains consistent when we resample the probes and assign checkpoints to phases separately for each seed. When we permute the probe identities between checkpoints, the higher exact fraction observed after grokking disappears. Lastly, the method recovers the expected components on fields with known Hodge decompositions.

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1 Introduction

A transformer trained to perform modular addition can generalise long after it has overfitted its training data [15], giving us a transition period in which we can track changes in the learned representation. We apply Hodge decomposition to the displacement field between successive activation checkpoints, separating representational motion into exact (gradient-like), coexact (curl-like), and harmonic components. This decomposition can produce an intuitive picture of what a representation is "doing." For modular addition, the successful solution also has an independently understood Fourier structure [13]. This lets us track how our geometric measurements change as Fourier structure forms in the activations. Using this reference, the baseline run suggests that grokking coincides with a shift from coexact-led to exact-led representational motion. Across additional runs, however, only the relative shift towards exact motion remains consistent, with the largest component before and after the transition varying between runs. Indeed, the difference between the coexact and exact fractions decreases from the grokking transition to the post-grokking phase in every tested seed and measurement setting. Since the velocity field depends on matching the same probes across checkpoints, we use a permutation null that shuffles the probe identities between checkpoints. This tests whether the effect reflects matched representational motion rather than the geometry of the two point clouds considered separately.

The network first fits the training set without generalising, before gradually developing a compact algorithmic solution. For modular addition $(a + b) \bmod P$ with P prime, Nanda et al. [13] used mechanistic interpretability to show that the generalising solution is a Fourier representation. The solution learned embeds each residue class $a \in \mathbb{Z}/P\mathbb{Z}$ as a point on a circle in a 2D subspace, with the angular position of a being $2\pi ka/P$ for some frequency k. Addition then follows from the trigonometric identity

$$\cos\!\left(\tfrac{2\pi k(a+b)}{P}\right) = \cos\!\left(\tfrac{2\pi ka}{P}\right)\cos\!\left(\tfrac{2\pi kb}{P}\right) - \sin\!\left(\tfrac{2\pi ka}{P}\right)\sin\!\left(\tfrac{2\pi kb}{P}\right). \tag{1}$$

This clock algorithm needs only a small number of Fourier frequencies. It places the P residue classes on a low-dimensional manifold with the topology of a circle for one frequency or a torus for several.

We also track changes in the representation's effective dimension and diffusion geometry. We check whether the Hodge result persists when we resample the probe inputs and change how we define the grokking period. We also verify the method using fields with known Hodge decompositions. Appendix B summarises the main results, the checks supporting them, and the limitations that remain.

2 Related Work

Existing work on grokking has approached the transition primarily through mechanistic interpretability and learning dynamics. We extend this work by studying the geometric reorganisation of activation space. On the mechanistic side, Nanda et al. [13] for example showed that the network learns a Fourier representation of modular addition. On the dynamics side, Liu et al. [12] extended grokking beyond algorithmic data, and Thilak et al. [16] linked grokking to late-training Slingshot dynamics under adaptive optimisation. Both lines of work motivate our perspective, which tracks how the geometry of the representation itself changes during the transition.

Recent work on sequence models suggests that memorised relational structure can take a geometric form [14]. This is relevant to grokking because the modular addition model does not merely memorise examples; after training it represents the cyclic group structure through Fourier-like coordinates [13].

To analyse representational motion, we use DiffusionGeometry [10, 11], a software package for applying differential geometry to point clouds. The package lets us represent the displacement of each activation as a differential one-form and decompose the resulting field into exact, coexact, and harmonic components.

DiffusionGeometry builds on spectral exterior calculus [4], which approximates differential forms and their operators from sampled data. It also uses diffusion geometry derived from a graph Laplacian. Diffusion maps [7] and Laplacian Eigenmaps [2, 3] provide the foundation for treating these graph-based operators as approximations to geometry on an underlying manifold.

More broadly, methodological work on interpretability asks whether an evaluation actually tests the claim attached to an explanation of model behaviour; in our case, the explanation concerns changes

in the geometry of the learned representation. Doshi-Velez and Kim [8] distinguish application-grounded, human-grounded, and functionally grounded evaluations. Adebayo et al. [1] show that saliency maps can look convincing even when they are insensitive to model parameters or training data. Hooker et al. [9] test feature attributions by removing the selected features, retraining the model, and measuring the change in performance. We test our geometric claim by repeating the analysis across independently trained models, varying reasonable analysis choices, and using a null that breaks point correspondence.

3 Experimental Setup

3.1 Architecture and Training

The training setup follows Nanda et al. [13] closely, with the single-head reduction discussed below; we record the parameters here for completeness.

We use a deliberately simplified single-head variant of the Nanda et al. [13] architecture:

- Task: modular addition $(a + b) \bmod 113$, with $P = 113$ (prime).
- Architecture: 1-layer transformer (TransformerLens 2.17.0), embedding dimension $d_{\text{model}} = 128$, single attention head ($d_{\text{head}} = 128$), MLP hidden dimension 512, no layer normalisation.
- Optimiser: AdamW with learning rate 10^{-3}, $\beta = (0.9, 0.98)$, and weight decay 1.0.
- Training: 25,000 full-batch epochs using cross-entropy loss.
- Data split: 70% of all (a, b) pairs held out for validation.
- Reproducibility: for each of eight runs, the data seed is also supplied to `torch.manual_seed` before model creation; there is no separate model seed. The runs use seeds 598, 599, 777, 1001, 2027, 3141, 4096, and 8080, and are pinned to TransformerLens 2.17.0.

We intentionally did not aim for an exact reproduction of the four-head model in Nanda et al. [13]. We reduced to a single head because preliminary four-head runs gave a substantially noisier circular-coordinate diagnostic, which made the later Fourier analysis difficult. The 1-head variant shows the same memorisation, transition, and generalisation pattern that we want to analyse, and we found its Fourier geometry easier to read. The circular-coordinate analysis remains supplementary and is not part of the eight-seed robustness claim.

At 51 checkpoints (every 500 epochs plus initialisation) we extract the MLP hidden activations (post-ReLU, $n_{\text{hidden}} = 512$) at the "=" token position for a fixed probe set of $n = 500$ subsampled input pairs. This gives a sequence of point clouds $X_t \in \mathbb{R}^{500\times512}$.

3.2 Methods

Spatial side: DG velocity Hodge decomposition. Given consecutive checkpoints X_t and $X_{t+\Delta t}$, the probe displacement field is

$$v_i(t) = x_i(t + \Delta t) - x_i(t).$$

We reduce X_t to 10 dimensions with PCA. We also project v_i through the same basis. We then build a DiffusionGeometry 0.1.0 model (commit `f5dc795`) [10, 11] with kNN $= 15$ and 50 spectral basis functions on the reduced point cloud. Let $V_t \in \mathbb{R}^{n\times d}$ denote the PCA-projected displacement and let $\{\phi_\ell\}$ denote the truncated diffusion function basis. Each row $V_{t,i}$ is a displacement vector in the Euclidean PCA space, so we identify it with the pointwise covector $\omega_i = \sum_{a=1}^{d} V_{t,ia}\,\mathrm{d}x^a$. The call `model.form(`$V_t$`, degree=1)` projects this covector field into the tensor-product basis $\{\phi_\ell\,\mathrm{d}x^a\}$. For each coordinate a, the coefficients c_a solve the measure-weighted projection $Gc_a = \Phi^\top M V_{t,:,a}$, where M is the empirical diffusion measure and $G = \Phi^\top M\Phi$ is the Gram matrix of the function basis. This construction represents displacement at each sampled point, rather than defining it as a difference along the edges of a graph. We then apply the Hodge decomposition

$$\omega = \mathrm{d}f + \delta g + h,$$

where $\mathrm{d}f$ is exact (the gradient of a 0-form), δg is coexact (the codifferential of a 2-form), and h is harmonic. The three components are orthogonal on a smooth manifold. On a finite point cloud the numerical cross terms are small but non-zero. The raw magnitude charts are noisy, so to interpret them we look at the composition rather than the raw magnitudes. For each component $c \in \{\mathrm{exact}, \mathrm{coexact}, \mathrm{harmonic}\}$, we calculate its normalised fraction as

$$\hat{e}_c = \frac{\|\omega_c\|^2}{\|\omega_{\mathrm{exact}}\|^2 + \|\omega_{\mathrm{coexact}}\|^2 + \|\omega_{\mathrm{harmonic}}\|^2}.$$

These three fractions sum to one. For example, a coexact fraction of 0.45 means that the coexact component contains 45% of the decomposed motion energy. We normalise by the sum of the component energies because the components have small non-zero cross terms on the finite point cloud. We interpret the harmonic term conservatively: without an independent topological analysis, such as persistent homology [6], we cannot treat harmonic energy as direct evidence of a topological cycle.

We test robustness across eight independently trained seeds and four random probe subsets of 400 examples. The code selects three approximately evenly spaced checkpoint pairs from each fixed phase: $0 \to 500$, $500 \to 1{,}000$, $1{,}000 \to 1{,}500$, $1{,}500 \to 2{,}000$, $2{,}500 \to 3{,}000$, $3{,}500 \to 4{,}000$, $4{,}000 \to 4{,}500$, $14{,}000 \to 14{,}500$, and $24{,}500 \to 25{,}000$. Our baseline uses 10 PCA dimensions, kNN $= 15$, and 50 basis functions. We then vary one choice at a time, using 5 or 20 PCA dimensions, kNN values of 10 or 20, and 30 or 75 basis functions. Across the seven settings and eight seeds, this produces 56 seed-by-setting results. Because results from the same seed are correlated, the eight training seeds form the independent units for inference. In the primary analysis, we classify pair midpoints before epoch 1,500 as pre-grokking, those from epochs 1,500 to 4,000 inclusive as the transition, and later midpoints as post-grokking. We also test a seed-specific definition anchored at the first saved checkpoint where validation accuracy reaches 0.5. For this definition, the transition extends from 1,500 epochs before to 1,000 epochs after that checkpoint. Reclassifying the same nine pairs gives between two and five pairs per phase, depending on the seed. The probe subsets use seeds 1101, 2202, 3303, and 4404. The same seeded generators draw one destination permutation for each checkpoint pair. They are reinitialised for each training seed, so 36 permutation index arrays are applied across eight models, producing $8 \times 4 \times 9 = 288$ permuted-field comparisons. We pair each observed result with its corresponding null result, average these comparisons within each seed, and then analyse the eight seed-level means. We calculate all reported 95% intervals as two-sided Student t intervals with 7 degrees of freedom. Finally, we test the implementation on gradient, rotational, and discrete harmonic fields with known Hodge decompositions.

Operator side: resolvent BW distance. At each checkpoint we build the symmetric normalised graph Laplacian L_t (adaptive-bandwidth Gaussian kernel, kNN $= 15$) on the PCA-reduced probe cloud [7] and extract $K = 30$ non-trivial eigenpairs, excluding the constant mode. The resolvent covariance is $\Sigma_t = (L_t + \varepsilon I)^{-1}$ on this retained non-trivial spectrum, with $\varepsilon = 0.01$, which maps small non-trivial Laplacian eigenvalues to large covariance eigenvalues and so emphasises low-frequency structure. We compute the BW distance $d_{\mathrm{BW}}(\Sigma_t, \Sigma_{t+\Delta t})$ between consecutive checkpoints [5].

Computing the BW distance requires both resolvent matrices to be expressed in the same coordinate system. Because we use the same probe inputs at every checkpoint, the rows of each eigenvector matrix $Q_t \in \mathbb{R}^{n \times K}$ retain the same input ordering, even though we recompute the PCA projection and graph. We use the epoch-0 non-trivial eigenvectors $Q_{\mathrm{ref}} = Q_0$ as a fixed reference and project each checkpoint's resolvent into this basis. Defining $C_t = Q_{\mathrm{ref}}^\top Q_t$, we obtain $\widetilde{\Sigma}_t = C_t \,\mathrm{diag}((\lambda_{t,k} + \varepsilon)^{-1})\, C_t^\top + 10^{-10} I$. The small diagonal term ensures that the matrix is positive definite, and we symmetrise it numerically before calculating the distance. Each consecutive comparison therefore uses $K \times K$ matrices expressed in the same epoch-0 spectral basis. The known correspondence between probes allows us to place the resolvents in a shared basis. We therefore avoid comparing eigenvalue spectra alone or treating eigenvectors computed at different checkpoints as already aligned.

We compare consecutive resolvent covariances using the Bures–Wasserstein distance [5]:

$$d_{\mathrm{BW}}^2(\Sigma_0, \Sigma_1) = \mathrm{tr}(\Sigma_0) + \mathrm{tr}(\Sigma_1) - 2\ \mathrm{tr}\left[\left(\Sigma_0^{1/2} \Sigma_1 \Sigma_0^{1/2}\right)^{1/2}\right]. \tag{2}$$

For zero-mean Gaussian distributions, this equals the Wasserstein-2 distance. Here the covariance matrices encode the diffusion geometry at each checkpoint, so the distance measures how much that geometry changes between checkpoints.

3.3 Testing the Geometric Interpretation

The baseline visualisation in Figure 1 suggests predominantly coexact motion during the transition and predominantly exact motion after grokking. We test this pattern across eight independently trained models, four random probe subsets of 400 examples, seven analysis settings, and both fixed and seed-specific assignments of checkpoints to phases. We also test the narrower prediction that the difference between the coexact and exact fractions decreases from the transition to the post-grokking phase. This prediction would be weakened if the direction reversed, its seed-level interval included zero, or it remained after point correspondence was destroyed.

The measured fractions change with the PCA dimension, neighbourhood size, and number of spectral basis functions, and the largest component varies. However, the difference between the coexact and exact fractions decreases in every seed and analysis setting. All eight models reach 99% validation accuracy between epochs 3,500 and 5,500. These results remain limited to this architecture, modular-addition task, and the analysis and phase assignments we tested.

4 Results

4.1 Training Dynamics

By epoch 500 the full-batch training loss has fallen to 0.002. Validation accuracy remains low through epoch 1,500 (12.9%), then improvement accelerates, reaching 94.3% by epoch 3,000 and 100% by epoch 4,000.

Later analyses tie back to this timeline as their foundation. Training divides into three segments: *memorisation* occurs between epochs 0 and 1,500. The shift known as *grokking* unfolds from epoch 1,500 onward, lasting until around epoch 4,000. Beyond that point performance settles, which marks the *post-grokking* period. Large shifts in validation accuracy, along with changes in how representations are structured, define what we call a phase transition; we use "phase transition" descriptively, as an analogy of a thermodynamic phase transition.

Figure 4 in the appendix shows the three-seed event study, with each run aligned at its first saved checkpoint where validation accuracy is at least 0.5. We base the statistical conclusions below on the analysis of all eight seeds.

4.2 Baseline Run: Hodge Decomposition of Representational Motion

Figure 1 shows the diffusion-geometric Hodge decomposition of representational motion across all 50 consecutive checkpoint pairs in the baseline run. In this run, the balance among exact, coexact, and harmonic motion changes across the grokking phases as follows:

1. Memorisation phase (epochs 0–1,500): the velocity is coexact-dominated on average (about half of the normalised energy), with the remainder split between exact and harmonic components. Step 1,000→1,500 shows the strongest coexact fraction ($\sim$72%). This indicates curl-dominated rearrangement of the point cloud during memorisation. The initial step 0→500 still behaves differently, with elevated harmonic energy; this likely reflects large-scale compression from random initialisation and residual global structure in the harmonic component rather than direct topological evidence. The exact and harmonic fractions in this short pre-grokking window are more numerically sensitive to the scientific-software environment than the transition-to-post directional result.
2. Grokking transition (epochs 1,500–4,000): the exact component rises within the window as the Fourier structure emerges, but the phase remains coexact-leading on average (exact fraction $\sim$ 33%, coexact fraction $\sim$ 54%, harmonic fraction $\sim$ 13%). The transition is therefore a gradual rise of exact motion within a still-curl-dominated regime, with the actual flip happening only post-grokking in the baseline seed. Across eight seeds at the baseline setting, the mean transition exact, coexact, and harmonic fractions are 0.325, 0.450, and

0.225, respectively. The coexact fraction exceeds the exact fraction on average. However, its dominance during the transition varies across seeds and parameter choices.

3. Post-grokking (epochs 4,000+): the exact component leads on average (mean exact fraction $\sim 69\%$, coexact fraction $\sim 21\%$, harmonic fraction $\sim 10\%$). The representation continues to undergo directed refinement, while the harmonic term remains a residual global component of the finite DG decomposition rather than direct evidence for a topological cycle. Across the eight baseline runs, the mean post-grokking exact, coexact, and harmonic fractions are 0.473, 0.331, and 0.196, respectively. The exact fraction is larger than the coexact fraction in six of the eight baseline runs and in 44 of the 56 seed-by-setting comparisons. Hence, the consistent result is a shift towards exact motion.

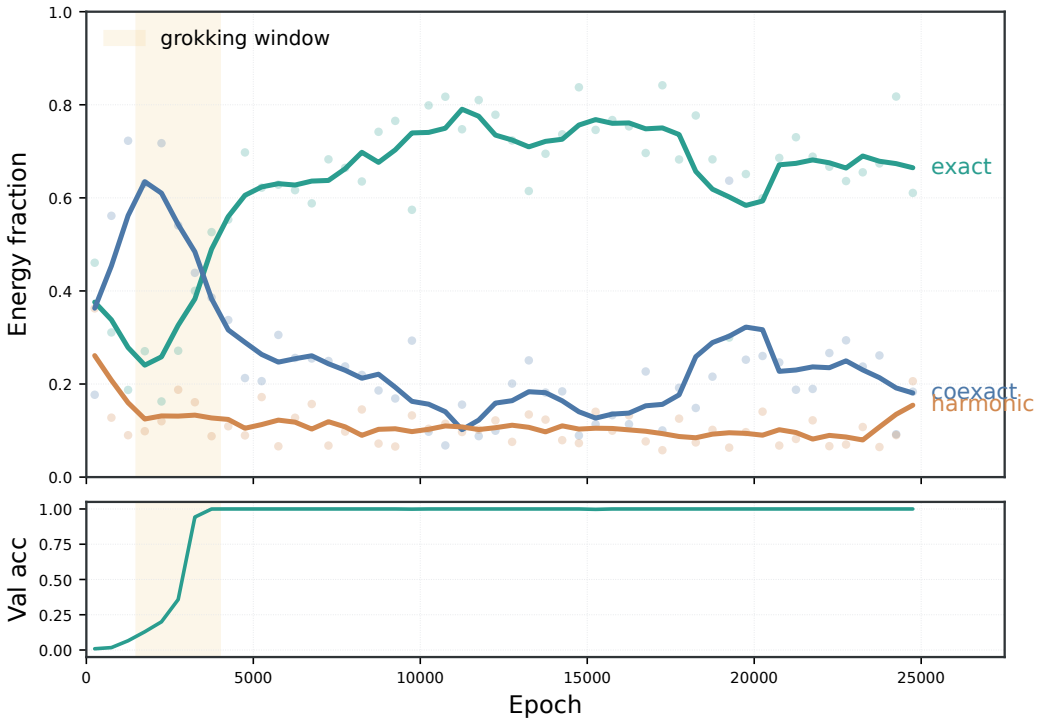


Figure 1: In the top panel, the translucent points show the exact, coexact, and harmonic fractions for individual checkpoint pairs. The solid lines are five-pair moving averages included to make the overall trend easier to see. Numerical values reported in the text refer to the individual points rather than the moving averages. The bottom panel shows validation accuracy. In this baseline run, the displacement field shifts from curl-like towards more gradient-like motion across grokking.

4.2.1 Results Across Training Seeds

Table 1 summarises the baseline Hodge fractions across the eight runs. Each training seed contributes one independent value.

Table 1: Baseline Hodge fractions across eight training seeds. For each phase, we average over the four probe subsets and selected checkpoint pairs within each seed. The table reports the mean of the resulting eight seed values.

Phase	Exact fraction	Coexact fraction	Harmonic fraction
Pre-grokking	0.342	0.488	0.170
Transition	0.325	0.450	0.225
Post-grokking	0.473	0.331	0.196

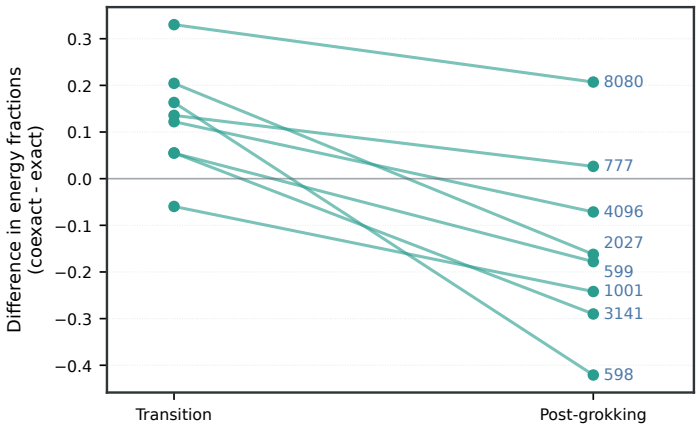


Figure 2: Difference between the coexact and exact fractions during the transition and post-grokking phases for each of the eight training seeds. Each line connects the two phase values for one seed after averaging over the four probe subsets and selected checkpoint pairs. All eight lines fall from the transition to the post-grokking phase. The different starting points and slopes show variation across seeds. Statistical inference uses the eight paired seed values.

Figure 2 shows the main result that remains consistent across seeds: the difference between the coexact and exact fractions decreases from the transition to the post-grokking phase in every run, although its value in each phase and the size of the decrease vary.

Across the eight paired seed-level values at the baseline setting, the mean change in $\hat{e}_{\text{coexact}} - \hat{e}_{\text{exact}}$ from the transition to the post-grokking phase was -0.267, with a 95% Student t interval of $[-0.400, -0.135]$.

For the correspondence null, we use 288 permuted-field comparisons: one for every combination of training seed, probe subset, and checkpoint pair at the baseline setting. We compare each observed field with its corresponding permuted field, then average the differences within each seed. After grokking, the exact fraction is on average 0.249 higher than under the null, with a 95% t interval of $[0.155, 0.344]$. The coexact fraction is 0.208 lower, with an observed-minus-null interval of $[-0.311, -0.104]$. When we permute the probe identities between checkpoints, the difference between the coexact and exact fractions changes very little from the transition to the post-grokking phase. Across the eight seeds, the mean change was 0.014, with a 95% Student t interval of $[-0.090, 0.118]$. The correctly matched fields showed a 0.281 larger decrease than the permuted null, with a 95% Student t interval of $[0.141, 0.421]$.

On fields with known Hodge decompositions, the method assigns 93.2% of the gradient field to the exact component, 98.8% of the rotational field to the coexact component, and approximately 94% of the discrete harmonic field to the harmonic component. The results also vary little across probe subsets. Across the tested settings, the median standard deviations between subsets are 0.015 for the exact fraction, 0.020 for the coexact fraction, and 0.015 for the harmonic fraction.

4.3 Additional Measures of Representational Change

4.3.1 Effective Dimensionality

From the 500×512 activation matrix, prior to any PCA compression, we form the centred activation covariance C_t (we do this for the same reason centring is done in PCA) and derive the effective dimension as the participation ratio

$$d_{\text{eff}}(t) = \frac{(\operatorname{tr} C_t)^2}{\operatorname{tr}(C_t^2)}.$$

In other words, the ratio is the participation ratio of the covariance eigenvalues. After starting at 50.7, d_{eff} quickly decreases and is already down to $\sim$24 by epoch 500. It reaches about 10 by the end of the grokking transition. Afterward, values lie mostly between 5 and 7, settling at ~ 6.4 by the

final checkpoint (Figure 3). The final representation is low-dimensional. The supervised diagnostic separately shows that Fourier information is recoverable.

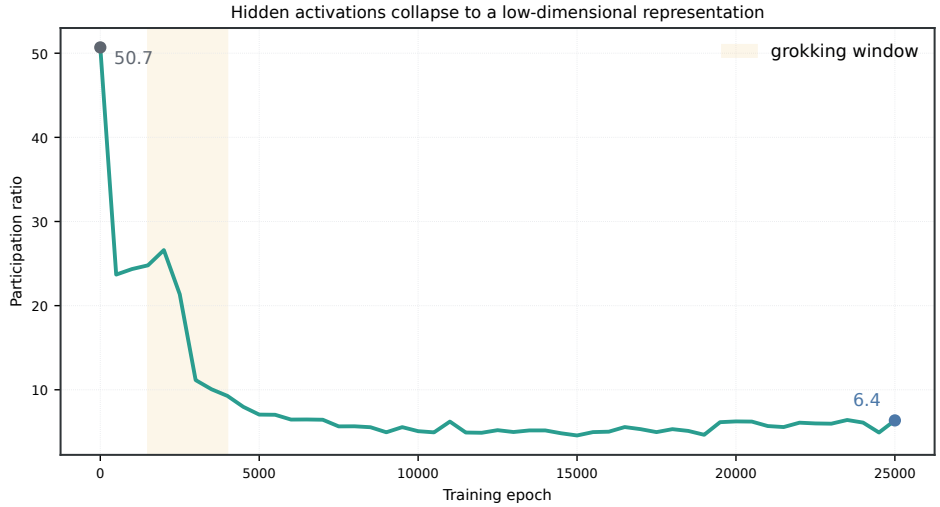


Figure 3: The hidden representation collapses from a diffuse high-dimensional cloud to a low-dimensional solution. The shaded window marks grokking; the curve shows the participation ratio falling from 50.7 at initialisation to ~ 6.4 by the final checkpoint. The earliest compression is largest, so this figure only shows dimensional collapse in general and is not yet a complete transition marker.

4.3.2 BW Distance Between Geometric States

Figure 4 in the appendix shows the resolvent BW distance across three seeds aligned at the first checkpoint where validation accuracy reaches 0.5. In the baseline run, the distance time series has three distinct features. The largest peak sits right at the start ($d_{\mathrm{BW}} = 7.34$ at epochs 0–500), as the diffuse initial geometry gives way to learned structure. A second, smaller spike appears at the grokking transition ($d_{\mathrm{BW}} = 1.36$ at 2,000–2,500, then 1.65 at 2,500–3,000). After that the consecutive distance settles into a much lower band, with mean $\bar{d}_{\mathrm{BW}} \approx 0.56$ and occasional fluctuations up to 1.23.

5 Discussion

5.1 What Remains Consistent Across Tests

The baseline run suggests that motion is led by the coexact component before and during grokking, then by the exact component afterwards. Across additional runs, however, this pattern is not universal. The exact fraction exceeds the coexact fraction after grokking in only 44 of the 56 combinations under the fixed phase assignment, and the relationship during the transition also varies. What remains consistent is a relative shift towards exact motion. The difference between the coexact and exact fractions decreases from the transition to the post-grokking phase in every seed and analysis setting under both ways of assigning checkpoint pairs to phases. At the baseline setting, the 95% interval for this change excludes zero. When point correspondence is destroyed, the higher exact fraction observed after grokking disappears. Together, these results indicate that the signal depends on tracking the same probe inputs between checkpoints.

We calculate the confidence intervals across eight independently trained models. The 56 results reuse each model across seven analysis settings, so they test sensitivity to those settings rather than provide additional independent observations. Resampling the probes tests sensitivity to the input sample, while the fields with known Hodge decompositions test whether the implementation recovers the expected components.

5.2 Grokking as Geometric Memory

The grokking experiment is a concrete case study for geometric memory [14]. During geometric algorithmisation, the model shifts from storing individual training examples to encoding the generalising algorithmic solution in a compact geometric configuration: the multi-frequency Fourier arrangement studied here and by Nanda et al. [13]. The third and final phase is geometric refinement. The algorithmic geometry has settled, the effective dimensionality plateaus at a low value, the BW distances drop into a low band, and the velocity is exact-dominated in the baseline seed. At this stage, the network fine-tunes its Fourier embedding after the earlier rotational reorganisation.

These observations suggest a possible mechanism. During grokking, the model may reorganise its representational coordinates as Fourier features displace memorisation-based features. This reorganisation could produce the relative coexact-like motion we observe. Once the Fourier coordinate system stabilises, continued optimisation may mainly refine the compact Fourier solution, producing a relative shift towards exact-like motion. Note this explanation does not predict that the coexact component must be the largest during every transition or that the exact component must be the largest afterwards. Although optimisation follows gradient descent in parameter space, the resulting activation velocity $\dot{X} = -(\mathrm{D}_\theta X)\nabla_\theta L$ need not be a gradient field in representation space because the representation Jacobian and Fourier subspace can change during training.

This mechanism remains a hypothesis with several testable predictions. We can decompose each checkpoint displacement into components that lie within and outside the final Fourier subspace, without changing the model's activations or predictions. If the hypothesis is correct, the post-grokking shift towards exact motion should be concentrated in the Fourier-aligned component. The coexact fraction should also track changes in the orientation of the Fourier subspace across checkpoints. We can further test whether the representation Jacobian or kernel stabilises before the Hodge shift, and whether changing weight decay alters the shift from rotational reorganisation to directed refinement.

6 Conclusion

The baseline run suggests a curl-like transition followed by gradient-like refinement. However, models trained with other seeds showed different compositions of exact, coexact, and harmonic motion. Despite this variation, the difference between the coexact and exact fractions decreases across all 56 combinations of eight seeds and seven analysis settings under both phase assignments. Tests on fields with known Hodge decompositions show that the method recovers the expected components, while the correspondence null shows that the result depends on matching the same probes across checkpoints. The evidence therefore supports a relative shift towards exact motion. Claims about a universally leading component or the mechanism behind the shift require further evidence.

A Additional Geometric Results

A.1 Construction of the Hodge Test Fields

We check the implementation using two vector fields on a regular planar grid and a discrete harmonic field on an annulus. For the exact and coexact checks, we sample 25 points from a 5×5 grid over $[-1, 1]^2$. The two fields are $v_{\text{exact}}(x, y) = (x, y)$, which is the gradient of $(x^2 + y^2)/2$, and $v_{\text{coexact}}(x, y) = (-y, x)$. We decompose both fields using two PCA dimensions, kNN $= 8$, and 20 basis functions. For the harmonic check, we sample 72 points from three circles with radii 0.6, 0.8, and 1.0, using 24 equally spaced angles on each circle. We project the angular field $(-y/r^2, x/r^2)$ onto the harmonic subspace of the fitted discrete complex, then decompose the resulting field using two PCA dimensions, kNN $= 8$, and 30 basis functions. The reported recovery fractions—93.2% exact, 98.8% coexact, and 94.2% harmonic—come from one deterministic run in the pinned DiffusionGeometry environment.

A.2 Three-Seed Event Study

For this analysis, we align each run at the first saved checkpoint where validation accuracy reaches 0.5. Figure 4 shows effective dimension, validation accuracy, BW distances, and the coexact fraction relative to this checkpoint. These three runs provide an exploratory view of when the geometric changes occur. The statistical analysis in the main text uses all eight seeds.

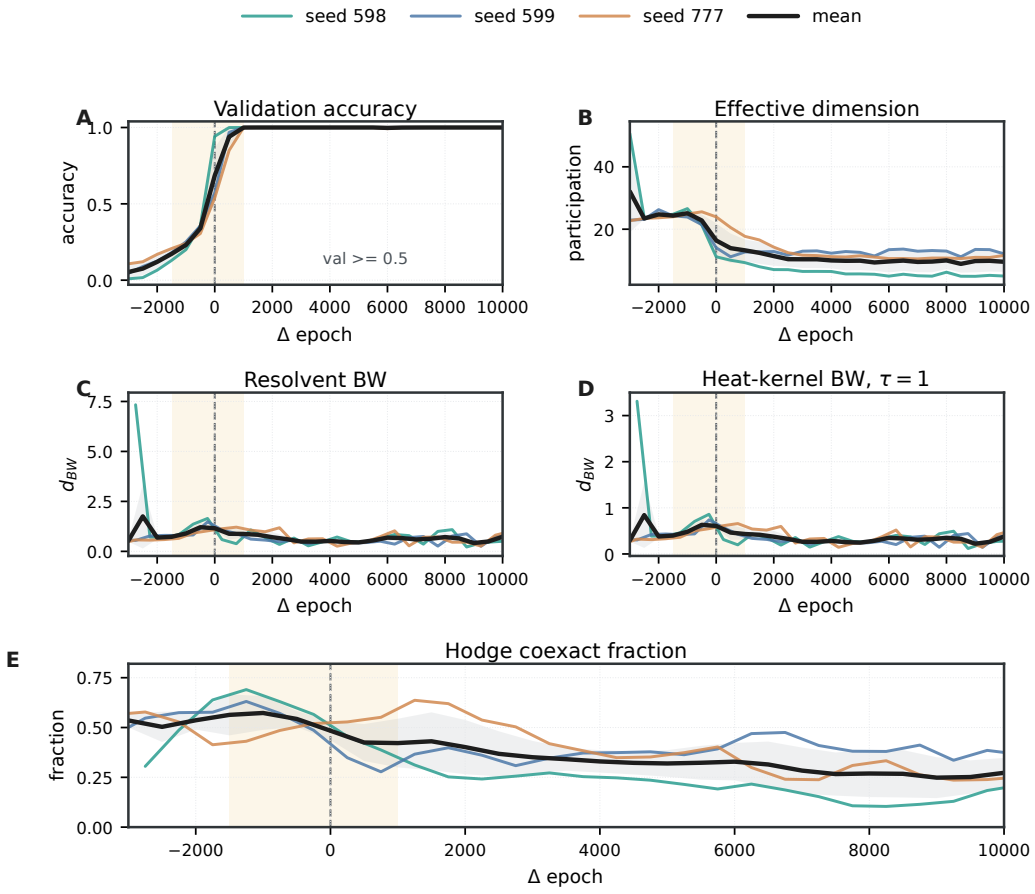


Figure 4: Three-seed event study aligned at the first saved checkpoint with validation accuracy of at least 0.5. Panel A shows validation accuracy, Panel B effective dimension, Panel C consecutive resolvent BW distance, Panel D consecutive heat-kernel BW distance at $\tau = 1$, and Panel E the Hodge coexact fraction.

A.3 Decomposed Velocity Fields

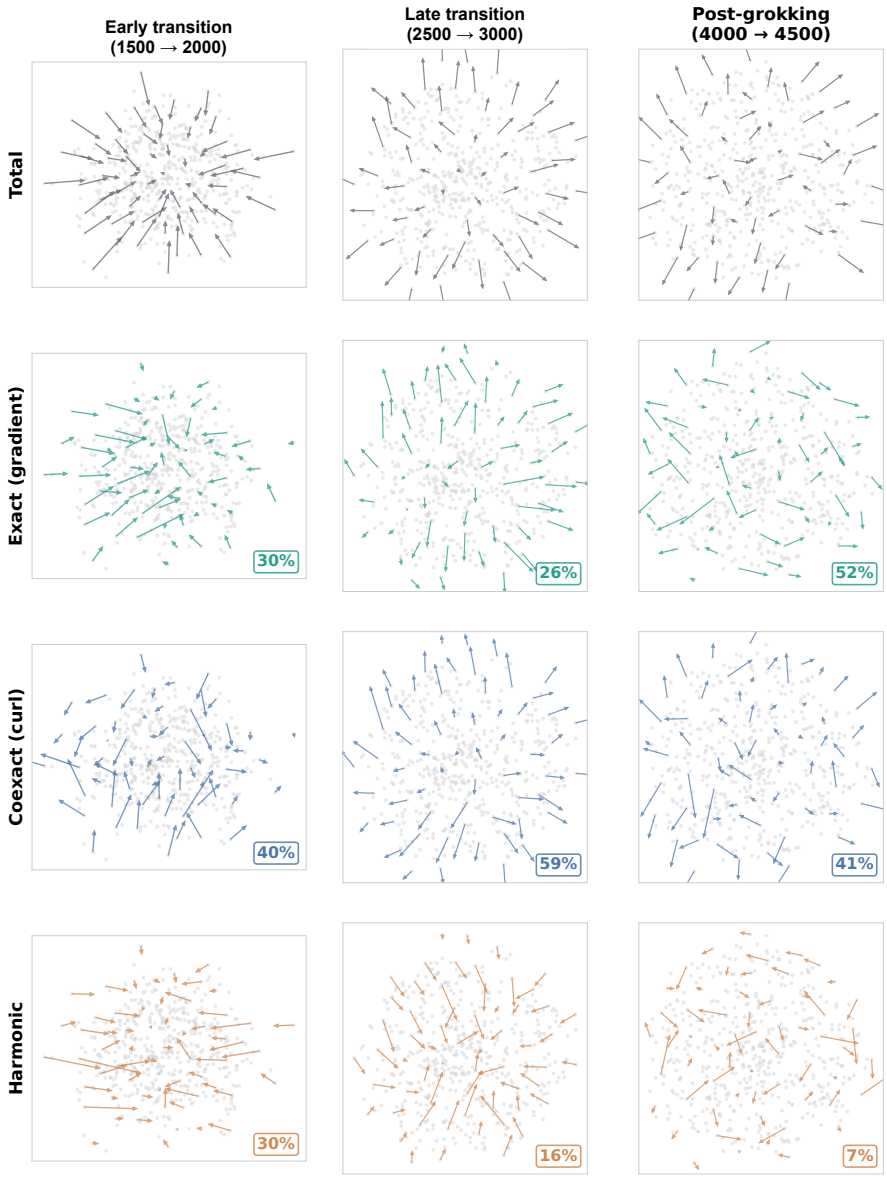


Figure 5: Decomposed velocity fields during grokking. Columns: early transition (1,500–2,000), late transition (2,500–3,000), and post-grokking (4,000–4,500). Rows: total velocity, exact (gradient) component, coexact (curl) component, and harmonic component. The plotted points and arrows are projected onto the first two principal components of the source checkpoint. To keep the figure readable, we show arrows for 60 points chosen to cover the plotted cloud. Arrow lengths are rescaled separately within each panel and should not be compared across panels. Fractions are annotated.

A.4 Fourier-Supervised Diagnostic

To check whether the trained activations contain the expected Fourier structure, we fit ridge regressions from the 512-dimensional activations to $\cos(2\pi k\ell/P)$ and $\sin(2\pi k\ell/P)$, where $\ell = (a+b) \bmod P$. We use 500 probes and a ridge penalty of $\alpha = 10$. The resulting two-dimensional coordinates are supervised Fourier directions. Figure 6 shows the mode $k = 38$, selected by its final-checkpoint score. Its joint cos/sin R^2 rises from 0.70 at initialisation to 0.98 by epoch 3,000 and 0.994 at the final checkpoint. Because the regressions are fitted and evaluated on the same probes in the $n < d$

setting, the initialisation score is inflated by overfitting. Other diagnostics select $k = 36$ or $k = 49$, so we do not treat $k = 38$ as a unique or canonical frequency. We use this analysis only as a check that Fourier information is recoverable from the trained activations.

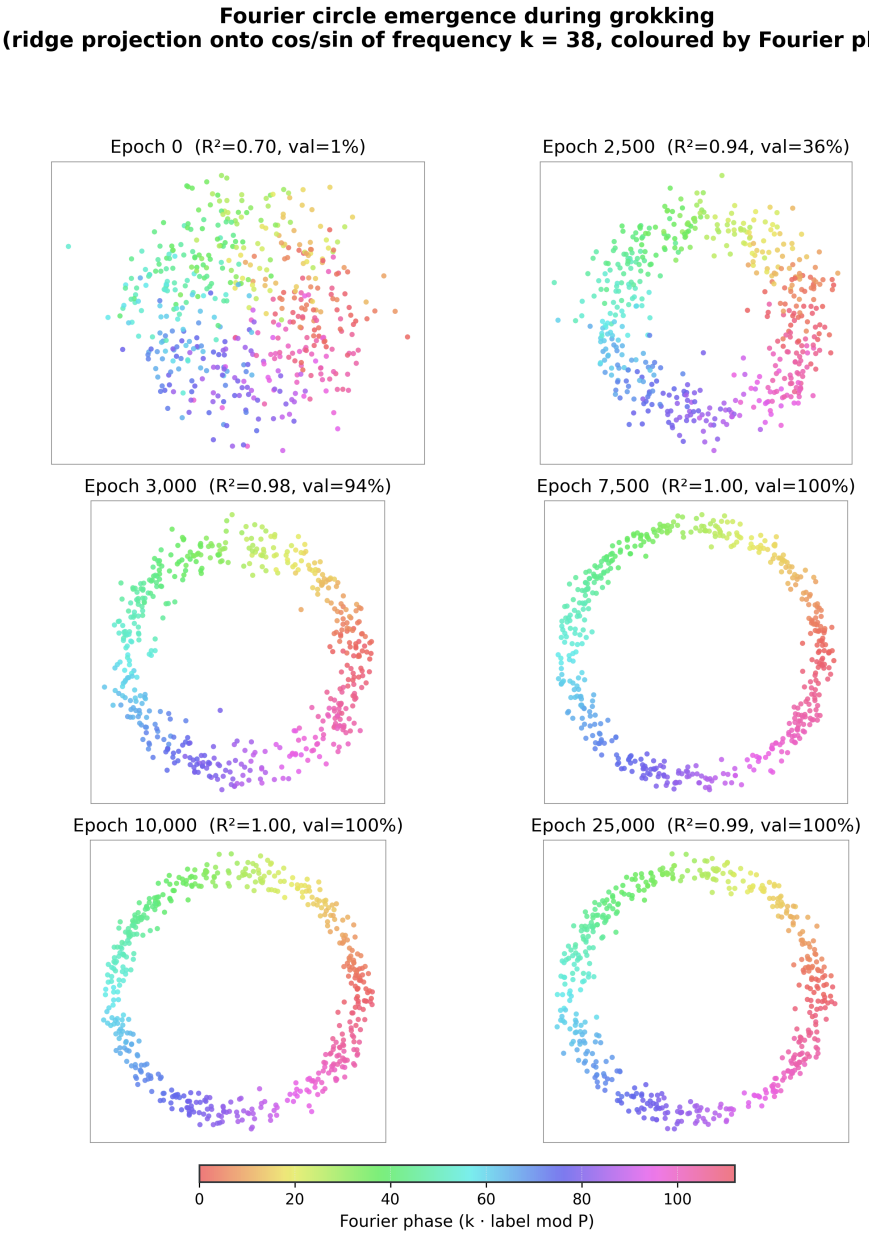


Figure 6: Fourier-supervised projection of the MLP hidden activations for frequency $k = 38$. The horizontal and vertical coordinates come from the cosine and sine ridge regressions. Points are coloured by Fourier phase. The projection is noisy at initialisation despite an inflated in-sample R^2 of 0.70. An ordered circle is visible by epoch 3,000, where $R^2 = 0.98$, and remains at the final checkpoint, where $R^2 = 0.994$. This supervised diagnostic shows that Fourier information is recoverable from the activations.

A.5 Heat-Kernel Distances Across Diffusion Scales

The complementary operator-level view replaces the resolvent covariance with the heat kernel

$$H_t(\tau) = e^{-\tau L_t},$$

evaluated at three scales $\tau \in \{0.1, 1.0, 10.0\}$ (local, mesoscopic, global). For consistency with the resolvent analysis, the heat-kernel BW distances are computed on the same retained non-trivial spectral subspace unless otherwise stated. The scale parameter sets the diffusion time at which the kernel is read out, so each scale exposes a different rate of geometric reorganisation.

At different scales (τ set to 0.1, 1.0 and 10.0) the heat kernel delivers a clearer picture than the resolvent. While the resolvent captures broad trends, the heat kernel separates changes at distinct diffusion times. Right when training begins, the BW distance spikes at all scales and then quickly falls once that phase ends. The peak distances are 3.77 at the local scale, 3.30 at the mesoscopic scale, and 1.13 at the global scale.

Indeed the global peak is roughly one-third the local peak. Another small peak shows up across each scale near epochs 2,000 to 3,000, matching the shift into grokking. Post-grokking, the three curves settle to low baseline fluctuations.

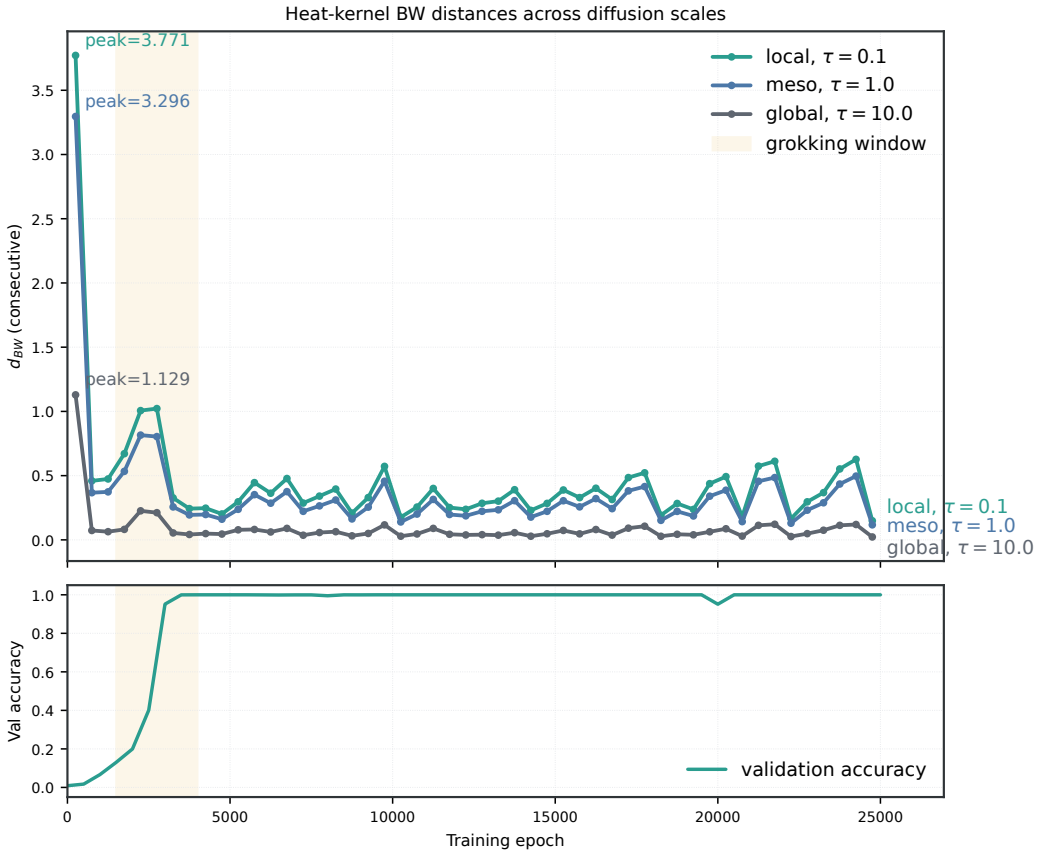


Figure 7: Heat-kernel BW shows the same operator change at the grokking transition across diffusion scales with the strongest locally and the weakest globally. The curves are consecutive BW distances between heat kernels at $\tau = 0.1, 1.0, 10.0$, with validation accuracy below. The shared shaded window makes the scale-specific increase comparable to the much larger initial restructuring.

B Claims, Evidence, and Limitations

Table 2 also identifies what would weaken our interpretation. The analysis on the operator side held under three independent training seeds for the dimensionality collapse and the BW spikes during the grokking transition. However, it could still lose its persuasiveness if those effects disappeared under held-out probe resampling. The difference between the coexact and exact fractions decreases consistently across eight seeds, resampled probes, and changes to the PCA dimension, neighbourhood

Table 2: Claims, evidence, robustness checks, and limitations for the grokking case study. Values written as mean $\pm$ standard deviation are calculated across the four probe subsets.

Claim	Primary evidence	Robustness or diagnostic check	Remaining limitation
Grokking has a sharp training transition	Validation accuracy remains low through $\sim$1,500 epochs, reaches 94.3% by 3,000, and reaches 100% by 4,000.	All geometric summaries are anchored to the same pre/transition/post timeline.	The timing varies across seeds: each of the eight runs reaches 99% validation accuracy between epochs 3,500 and 5,500.
Representational motion shifts towards the exact component after grokking.	At the baseline setting, the mean difference between the coexact and exact fractions decreases by 0.267 from the transition to the post-grokking phase.	We repeat the analysis across eight seeds, four probe subsets, seven settings, and seed-specific phase alignment. We also test the method on fields with known Hodge decompositions and use 288 paired permutations to test the importance of point correspondence.	The exact fraction exceeds the coexact fraction after grokking in 44 of the 56 combinations using fixed phase ranges. The phase ranges and analysis settings are not exhaustive.
Representation dimension collapses	Participation-ratio effective dimension falls from 50.7 at initialisation to 6.4 post-grokking.	Four 400-example probe subsets give $49.63 \pm 0.50 \to 6.40 \pm 0.03$, ratio 0.129 ± 0.002. Three seeds give final/initial 0.126, 0.194, 0.250.	Probe-subset and three-seed checks.
Operator geometry restructures	Resolvent BW distance has a large initial spike ($d_{\mathrm{BW}} = 7.34$), a transition-time burst (1.36, 1.65), and a lower post-grokking band.	Probe subsets preserve the pattern: initial 6.258 ± 0.050, transition peak 1.662 ± 0.057, post mean 0.531 ± 0.016, transition/post 3.13 ± 0.09. Three seeds give transition/post 2.92, 2.64, 1.48 (2.29, 1.84, 1.44 after accuracy alignment).	Graph, PCA, and reference-basis choices.
Local diffusion geometry changes at the transition	Heat-kernel BW curves show a secondary increase across local, meso, and global diffusion scales.	For probe subsets, transition/post ratios are 3.12 ± 0.11 ($\tau = 0.1$), 3.21 ± 0.11 ($\tau = 1$), and 3.36 ± 0.05 ($\tau = 10$); all four subsets have transition peak above post mean at every scale. Three seeds give $\tau = 1$ peak/post 2.99, 2.47, 1.52.	Only three diffusion scales are tested.
Fourier information becomes linearly recoverable	Fourier ridge projections and circular-coordinate scans recover post-grokking circular or multi-frequency structure.	The result is treated as an exploratory diagnostic.	In-sample ridge; best-pair circular scan; correlation varies in seeds 599 and 777.

size, and number of spectral basis functions. The circular-coordinate results remain supplementary because we have not tested them as extensively.

References

[1] Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems*, volume 31, 2018. URL https://proceedings.neurips.cc/paper/2018/hash/294a8ed24b1ad22ec2e7efea049b8737-Abstract.html.

[2] Mikhail Belkin and Partha Niyogi. Laplacian eigenmaps for dimensionality reduction and data representation. *Neural Computation*, 15(6):1373–1396, 2003. doi: 10.1162/089976603321780317.

[3] Mikhail Belkin and Partha Niyogi. Towards a theoretical foundation for Laplacian-based manifold methods. *Journal of Computer and System Sciences*, 74(8):1289–1308, 2008. doi: 10.1016/j.jcss.2007.08.006. Conference version: COLT 2005.

[4] Tyrus Berry and Dimitrios Giannakis. Spectral exterior calculus. *Communications on Pure and Applied Mathematics*, 73(4):689–770, 2020. doi: 10.1002/cpa.21885. arXiv:1802.01209.

[5] Rajendra Bhatia, Tanvi Jain, and Yongdo Lim. On the Bures–Wasserstein distance between positive definite matrices. *Expositiones Mathematicae*, 37(2):165–191, 2019. doi: 10.1016/j.exmath.2018.01.002.

[6] G. Carlsson. Topology and data. *Bulletin of the American Mathematical Society*, 46(2):255–308, 2009. doi: 10.1090/S0273-0979-09-01249-X.

[7] R. R. Coifman and S. Lafon. Diffusion maps. *Applied and Computational Harmonic Analysis*, 21(1):5–30, 2006. doi: 10.1016/j.acha.2006.04.006.

[8] Finale Doshi-Velez and Been Kim. Towards a rigorous science of interpretable machine learning. *arXiv preprint arXiv:1702.08608*, 2017. doi: 10.48550/arXiv.1702.08608.

[9] Sara Hooker, Dumitru Erhan, Pieter-Jan Kindermans, and Been Kim. A benchmark for interpretability methods in deep neural networks. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL https://proceedings.neurips.cc/paper/2019/hash/fe4b8556000d0f0cae99daa5c5c5a410-Abstract.html.

[10] Iolo Jones. Diffusion geometry. *arXiv preprint arXiv:2405.10858*, 2024. doi: 10.48550/arXiv.2405.10858.

[11] Iolo Jones and David Lanners. Computing diffusion geometry. *arXiv preprint arXiv:2602.06006*, 2026. doi: 10.48550/arXiv.2602.06006. Computational framework for diffusion geometry.

[12] Z. Liu, E. J. Michaud, and M. Tegmark. Omnigrok: Grokking beyond algorithmic data. In *International Conference on Learning Representations (ICLR)*, 2023. URL https://openreview.net/forum?id=zDiHoIWa0q1. arXiv:2210.01117.

[13] Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. In *International Conference on Learning Representations*, 2023. doi: 10.48550/arXiv.2301.05217. arXiv:2301.05217.

[14] Shahriar Noroozizadeh, Vaishnavh Nagarajan, Elan Rosenfeld, and Sanjiv Kumar. Deep sequence models tend to memorize geometrically; it is unclear why. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026. doi: 10.48550/arXiv.2510.26745. arXiv:2510.26745.

[15] A. Power, Y. Burda, H. Edwards, I. Babuschkin, and V. Misra. Grokking: Generalization beyond overfitting on small algorithmic datasets. *arXiv preprint arXiv:2201.02177*, 2022. doi: 10.48550/arXiv.2201.02177.

[16] Vimal Thilak, Etai Littwin, Shuangfei Zhai, Omid Saremi, Roni Paiss, and Joshua Susskind. The slingshot mechanism: An empirical study of adaptive optimizers and the grokking phenomenon. *arXiv preprint arXiv:2206.04817*, 2022. doi: 10.48550/arXiv.2206.04817.